Wally Ahmed

wally00.dev@gmail.com | (929) 266-3964 | Manhattan, NY | http://linkedin.com/in/wally-ahmed

SUMMARY

Rising software engineer and AI researcher with experience building **Agentic automation tools**, conducting applied **AI research**, and **developing data-driven solutions** for sustainability challenges. Experienced working on interdisciplinary research through The Grove School of Engineering at CCNY. **Building agentic workflows,** implementing **RAG systems,** creating **voice agents, and data analysis and research** to solve real-world problems. Passionate about artificial intelligence, software engineering, and using technology to create scalable environmental and societal impact. Seeking internships where I can contribute technical skills while continuing to grow as a software engineer.

TECHNICAL EXPERIENCE

Founder & Lead Engineer *OpenQuant-Agentic Trading Platform* **May 2026 – Present**

- Built a **multi-tenant agentic trading platform** in **TypeScript**, **Node.js**, **PostgreSQL**, and **React** where autonomous AI agents research live markets, plan, and execute trades through a gated **research, plan, execute, review** loop on real brokerage APIs. It has **paper trading**, **options support**, and a full **KYC**, account, and funding onboarding flow.
- Built a **model-agnostic agent harness** that works across commercial LLM gateways (**300+ models**) and self-hosted **Hugging Face** models with **LoRA adapters** on serverless **GPU** infrastructure, so users can bring their own fine-tuned trading models.
- Added **defense-in-depth safety** to the core with **prompt-injection fencing**, a **deny-by-default tool gateway**, **fail-closed trade guards**, strict **per-agent tenant isolation**, and kill-switches, so untrusted market data gets neutralized before it can reach a model or a trade path.
- Built a **per-agent semantic memory layer** that recalls each research cycle's lessons through **semantic search**, plus a **skills system** where agents write, version, and track how well their own trading strategies work behind human and automated **approval gates**.
- Backed the platform with **4,800+ automated tests**, **Stripe** billing, and a **React** dashboard that shows live portfolios and agent-memory search.

Technical Intern *Engineers Without Borders USA (CCNY Chapter)* **November 2025 – May 2026**

- Built agentic **no-code pipelines** that turn reports and stories into interactive generated blog pages, using **vision models** for **multi-modal data interpretation** to build the **UI/UX** for each new page.
- Moved the chapter codebase from static HTML to a fully interactive React/Next.js platform to improve maintainability and follow best practices.
- Used AI development tools like **Claude**, **Cursor**, **ChatGPT-CodeX**, **GitHub Copilot**, and **Microsoft Copilot** to speed up debugging, prompt engineering, and feature development.
- Handled ongoing site updates, maintenance, and deployments to keep the content and features current.

Software Engineering Fellow *Springboard - Software Engineering Career Track* **May 2023 – Jan 2025**

- Completed **800+ hours** of hands-on coursework with mentors covering the fundamentals of **Frontend**, **Backend**, **Fullstack development**, **Database Management**, and **Software Architecture**.
- Learned front-end and back-end web development from industry mentors with JavaScript, HTML, CSS, jQuery, Bootstrap, Python, Flask, Node.js, Express, SQL, PostgreSQL, RESTful APIs, and Git/GitHub, plus data structures, Problem-Solving, and Algorithm Design.
- Completed a 1-year intensive bootcamp in Software Engineering methods and technologies.

LEADERSHIP EXPERIENCE

Hermes Handoff | Hackathon Winner **July 2026**

- Placed **1st** in the **Cotal.ai sponsor track** at the **AGI Summit 2026 Hackathon** with a wizard CLI that finds which AI coding harnesses are installed and signed in on a machine, then routes any task (research, files, data, or code) through the user's existing subscription instead of paid API credits.
- Built **multi-agent mesh connectors** in **TypeScript** for **OpenAI Codex CLI** and **Google Antigravity**, so agents on different vendors' harnesses can take delegated work as autonomous workers.
- Designed a **12-task benchmark suite** (coding, agentic, recall, reasoning, hallucination resistance) and published **11 hermetic runs as 9 studies**, which showed harness-per-model routing beat single-loop multi-model delegation by **~30% wall-clock** at equal quality.
- Tested fault tolerance in a follow-up benchmark run by killing a worker mid-task, and the mesh caught the failure, reassigned the work, and finished in **122 seconds**. Documented the system as a **narrated 32-scene interactive walkthrough** deployed on **Fly.io**.

Live site: https://hermes-handoff-overview.fly.dev/

Flightdeck-AI Software Factory | Hackathon Winner **June 2026**

- Placed **1st in the Render Track** leading a team that built an AI-powered software delivery workflow with automated planning, coding, testing, review, and deployment stages.
- Led development of a gated pipeline that made preview deployments, so reviewers could test features before approving pull requests.
- Owned the technical stack, including Render deployment services, AI-agent workflows, React/Vite dashboards, and persistent run-state tracking for end-to-end software automation.

Live site: https://flightdeck-board.onrender.com/architecture

City Hacks The State | Hackathon Winner AI Facilities Intake & Work-Order Platform **June 2026**

- Placed **Top 5 overall** and won the **Best Architecture** award, leading the team that designed a scalable, AI-powered facilities-reporting system that takes requests through web chat, Telegram, and voice calls.
- Built an agentic workflow that verifies facility locations, classifies issue priority, drafts clear maintenance reports, and asks the user to confirm before submitting work orders.
- Connected the AI intake workflows to a work-order dashboard, **GraphQL** APIs, **MongoDB**, **Telegram**, **Twilio**, and server-side routes that kept sensitive credentials protected.
- Coordinated the team across intake, backend, and dashboard workstreams and pulled each piece into one demo that students could use to submit building issues end to end.

Live site: https://ca-intake-ae320e.fly.dev/architecture.html

Open Source Contributor **Ongoing**

- Authored **four MIT-licensed open-source MCP (Model Context Protocol)** projects, three **TypeScript** servers plus a one-command installer, that let any AI coding agent hand work to outside models as subagents through **OpenRouter** (any model, with reasoning-effort and token-budget control), **OpenAI GPT** (ChatGPT subscription OAuth or API key), and **Google Antigravity**.
- Added a reusable **orchestration-pattern library** to each server, so an agent pulls up a proven delegation workflow instead of improvising prompts. These servers were the subagent baselines in the Hermes Handoff benchmark study.
- Submitted **three pull requests** to Cotal.ai's open-source agent-orchestration monorepo, including a **Google Antigravity connector** and a connector for OpenAI's Codex CLI, each ported from a working prototype to the maintainers' **TypeScript** standards.
- Found and fixed a bug where crashed agents stayed listed as online indefinitely, submitting **five repairs** in one pull request (12 files, +687/−9) with **three new regression tests** covering the failure modes.
- Rebased the pull requests across four upstream releases, passing the project's full **typecheck**, **build**, and **CI smoke-test** gates on every rebase.

Link: https://github.com/Cotal-AI/Cotal/pulls?q=author%3AWally-Ahmed

SKILLS

Languages: TypeScript, JavaScript, Python, SQL, Bash/Shell
Frontend: React, Next.js, Vite, Tailwind CSS, HTML/CSS
Backend / Data: Node.js, Express, PostgreSQL, MongoDB, GraphQL, REST APIs, Docker
AI / Agents: LLM agent orchestration (multi-agent meshes, failover), MCP (Model Context Protocol) server development, agent benchmarking & evaluation, RAG, vector search (Pinecone), prompt-injection defense, Hugging Face fine-tuning (LoRA), voice agents (TTS/STT)
Cloud / DevOps: AWS, Microsoft Azure, Fly.io, Render, RunPod (serverless GPU), Stripe, Twilio, Git/GitHub, GitHub Actions, CI/CD, pnpm monorepos

CERTIFICATIONS

Google AI Essentials – Google **May 2026**
Digital Marketing Certified – HubSpot Academy **May 2026**

EDUCATION

Grove School of Engineering (CUNY) – Bachelor of Science (BS) in **Computer Science**